Sets

Jon Friskics

Senior Software Development Author, Pluralsight

@jonfriskics@mastodon.social | www.pluralsight.com

JavaScript Sets: Create a Set

```
let conditions = new Set(["Sunny","Rainy","Cloudy","Snowy"]);
```

conditions

value	Sunny	Rainy	Cloudy	Snowy

```
let set2 = new Set(["Sunny","Rainy","Rainy","Rainy","Rainy"]);
```

set2

value	Sunny	Rainy

JavaScript Sets: Set Data Is Not Sortable

```
let conditions = new Set(["Sunny","Rainy","Cloudy","Snowy"]);
```

Order matches the order data was added

conditions

value	Sunny	Rainy	Cloudy	Snowy

JavaScript Sets: Add, Delete, and Clear Elements

```
let conditions = new Set(["Sunny","Rainy","Cloudy","Snowy"]);

conditions.add("Clear");
```

conditions

value	Sunny	Rainy	Cloudy	Snowy	Clear

JavaScript Sets: Add, Delete, and Clear Elements

```
let conditions = new Set(["Sunny","Rainy","Cloudy","Snowy"]);

conditions.add("Clear");
conditions.delete("Sunny");
```

conditions

value	Sunny	Rainy	Cloudy	Snowy	Clear

JavaScript Sets: Add, Delete, and Clear Elements

```javascript
let conditions = new Set(["Sunny","Rainy","Cloudy","Snowy"]);

conditions.add("Clear");
conditions.delete("Sunny");
conditions.clear();
```

conditions

value	Rainy	Cloudy	Snowy	Clear

JavaScript Sets: Check For Data and Size

```javascript
let conditions = new Set(["Sunny","Rainy","Cloudy","Snowy"]);

if(conditions.has("Rainy")) {
  // do something with that info
};

let size = conditions.size // size == 4
```

conditions

value	Sunny	Rainy	Cloudy	Snowy

JavaScript Sets: Iteration with for...of

```javascript
let conditions = new Set(["Sunny","Rainy","Cloudy","Snowy"]);

for(let condition of conditions) {
  console.log(condition);
};


// logs each condition on a new line:
// Sunny
// Rainy
// Cloudy
// Snowy
```

JavaScript Sets: Iteration with forEach

```
let conditions = new Set(["Sunny","Rainy","Cloudy","Snowy"]);

conditions.forEach((condition) => {
  console.log(condition);
});

// also logs each condition on a new line:
// Sunny
// Rainy
// Cloudy
// Snowy
```

Comparing TypeScript Collections

Collection Name	Arrays	Tuples	Sets	Maps
Order type	Ordered and can reorder	Ordered by type position	Ordered by insertion order	
How to declare	type[] or Array<type>	[type1, type2]	new Set<type>()	
Index style	Positional index, starting at 0	Positional index, starting at 0	No index, lookup by value	
Duplicate values	Can contain duplicate values	Can contain duplicate values	Can not contain duplicate values	
Length	Any length, can grow and shrink	Fixed length	Any length, can grow and shrink	
Sortable	Sortable	Not sortable	Not sortable	
Mutable	Mutable by default	Mutable by default	Mutable by default	
Immutable version	readonly type[] or ReadonlyArray<type>()	readonly [type1, type2]	ReadonlySet<type>()	